



Locomotion and Movement

Movement is one of the significant features of living beings. Animals and plants exhibit a wide range of movements. The **streaming of protoplasm** in unicellular organisms like *Amoeba* is a simple form of movement; the movement of **cilia, flagella and tentacles** is shown by many organisms; and human beings can move their **limbs, jaws, eyelids, tongue**, etc.

- **Locomotion** - some movements result in a change of place or location. Such voluntary movements are called locomotion. Walking, running, climbing, flying and swimming are all forms of locomotory movements.
- Locomotory structures **need not be different** from those affecting other types of movements. For example, in *Paramecium*, cilia help in the movement of food through the cytopharynx *and* in locomotion as well; *Hydra* can use its tentacles for capturing prey *and* for locomotion; and we use our limbs for changes in body posture *and* for locomotion.

These observations suggest that movements and locomotion **cannot be studied separately**. The two may be linked by stating that **all locomotions are movements but all movements are not locomotions**.

- Methods of locomotion vary with the animal's **habitat** and the demand of the situation. However, locomotion is **generally** for the search of food, shelter, mate, suitable breeding grounds, favourable climatic conditions, or to escape from enemies/predators.

17.1 Types of Movement

Cells of the human body exhibit **three main types of movements**, namely, **amoeboid, ciliary and muscular**.

Type	Where it occurs	How it works
Amoeboid	Specialised cells like macrophages and leucocytes in blood	Effected by pseudopodia formed by the streaming of protoplasm (as in <i>Amoeba</i>); cytoskeletal elements like microfilaments are also involved.
Ciliary	Most internal tubular organs lined by ciliated epithelium	Coordinated cilia in the trachea help remove dust particles and foreign substances inhaled with air; the passage of ova through the female reproductive tract is also facilitated by ciliary movement.
Muscular	Limbs, jaws, tongue, etc.	Uses the contractile property of muscles ; effectively used for locomotion and other movements by human beings and the majority of multicellular organisms.

① **[NOTE]** Locomotion requires a **perfect coordinated activity of the muscular, skeletal and neural systems**. This chapter covers the types of muscles, their structure, the mechanism of their contraction, and the important aspects of the skeletal system.

17.2 Muscle

You have studied in Chapter 8 that **cilia and flagella are outgrowths of the cell membrane**. Flagellar movement helps in the **swimming of spermatozoa**, the maintenance of water current in the **canal system of sponges**, and in the locomotion of Protists like *Euglena*.

- **Muscle** - a specialised tissue of **mesodermal origin**. About **40-50 per cent** of the body weight of a human adult is contributed by muscles.
- Muscles have special properties: **excitability, contractility, extensibility and elasticity**.

Muscles have been classified using different criteria, namely **location, appearance and nature of regulation** of their activities. Based on their **location**, three types are identified: **(i) Skeletal (ii) Visceral and (iii) Cardiac**.

Muscle type	Location	Appearance	Regulation	Chief role
Skeletal	Closely associated with the skeletal components of the body	Striped / striated under the microscope	Under voluntary control of the nervous system (voluntary muscles)	Primarily locomotory actions and changes of body posture

Muscle type	Location	Appearance	Regulation	Chief role
Visceral	Inner walls of hollow visceral organs - alimentary canal, reproductive tract, etc.	No striation, smooth (smooth / nonstriated muscle)	Not under voluntary control (involuntary muscles)	Transport of food through the digestive tract and gametes through the genital tract
Cardiac	Muscles of the heart ; many cells assemble in a branching pattern	Striated	Involuntary - the nervous system does not control them directly	Contraction of the heart

Structure of a skeletal muscle. Each organised skeletal muscle is made of a number of **muscle bundles** or **fascicles** held together by a common collagenous connective tissue layer called **fascia**. Each muscle bundle contains a number of **muscle fibres** (Figure 17.1). The bundle is also served by a **blood capillary** supply drawn in the cross-section.

- **Sarcolemma / sarcoplasm.** Each muscle fibre is lined by the plasma membrane called **sarcolemma** enclosing the **sarcoplasm**. A muscle fibre is a **syncytium** (source spelling) as the sarcoplasm contains **many nuclei**.
- The endoplasmic reticulum, i.e. the **sarcoplasmic reticulum** of the muscle fibres, is the **store house of calcium ions** (Ca^{++}).
- A characteristic feature of the muscle fibre is a large number of parallelly arranged **filaments** in the sarcoplasm called **myofilaments** or **myofibrils**.

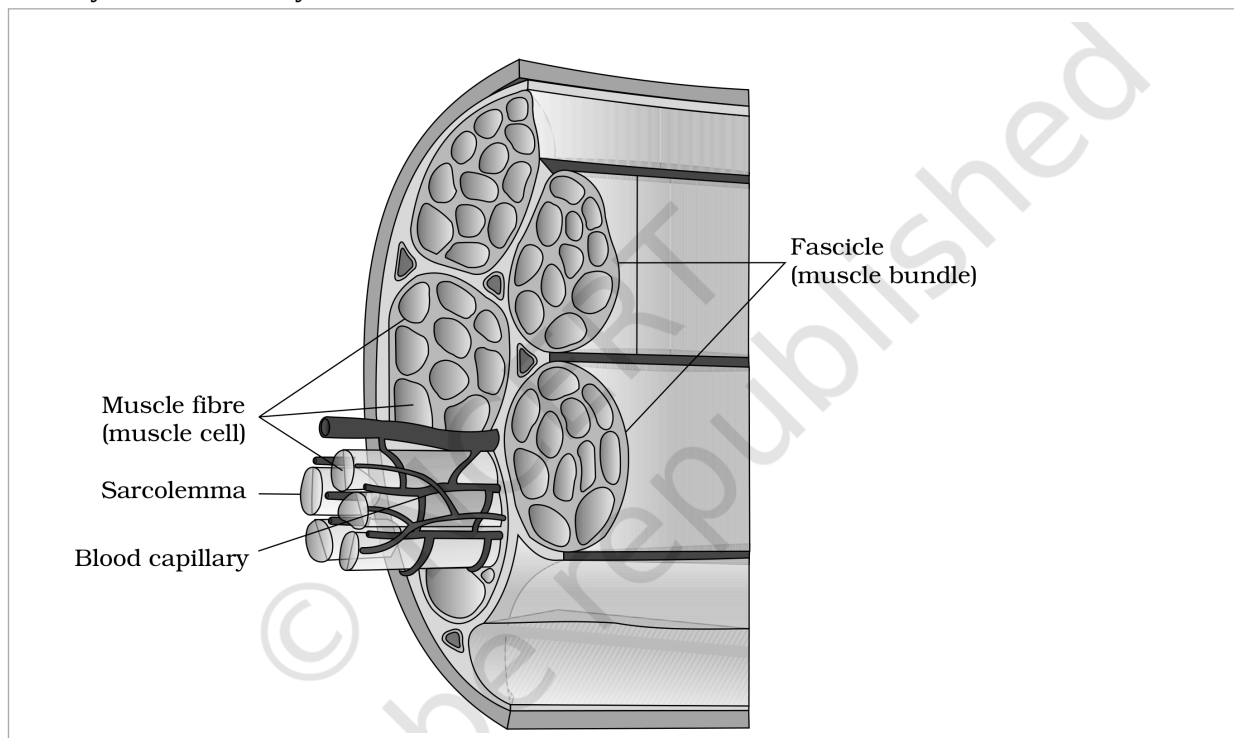


Figure 17.1 Diagrammatic cross sectional view of a muscle showing muscle bundles and muscle fibres

① [NOTE] **Figure 17.1 labels:** Fascicle (muscle bundle); Muscle fibre (muscle cell); Sarcolemma; Blood capillary.

Banding pattern. Each myofibril has **alternate dark and light bands** on it. The striated appearance is due to the distribution pattern of two important proteins - **Actin and Myosin**.

Band / line	Also called	Contents
'I' band (light)	Isotropic band	Contains actin (thin filament)
'A' band (dark)	Anisotropic band	Contains myosin (thick filament)
'Z' line	Elastic fibre bisecting each 'I' band	Thin filaments are firmly attached to it
'M' line	Thin fibrous membrane	Holds the thick filaments together in the middle of the 'A' band
'H' zone	Central part of the 'A' band	The part of the thick filament not overlapped by thin filaments - i.e. thick filaments only

- Both proteins are arranged as **rod-like structures**, parallel to each other and to the longitudinal axis of the myofibrils. **Actin filaments are thinner** than the myosin filaments, hence they are commonly called **thin** and **thick** filaments respectively.
- The '**A**' and '**I**' bands are arranged **alternately** throughout the length of the myofibrils.
- **Sarcomere** - the portion of the myofibril between **two successive 'Z' lines**. It is the **functional unit of contraction** (Figure 17.2). (The muscle **fibre** itself is the **anatomical unit** of muscle, while the sarcomere is its functional unit.)
- In a **resting state**, the edges of the thin filaments on either side of the thick filaments **partially overlap** the free ends of the thick filaments, leaving the central part of the thick filaments (the '**H**' zone) free.

☆ **[MEMORY AID - not in NCERT]** Each **sarcomere** = one central '**A**' band (thick myosin filaments) flanked by **two half 'I' bands** (thin actin filaments), the whole unit marked off by '**Z**' lines at both ends. Mnemonic for what shortens: only the '**I**' band and the '**H**' zone shrink on contraction - the '**A**' band never does.

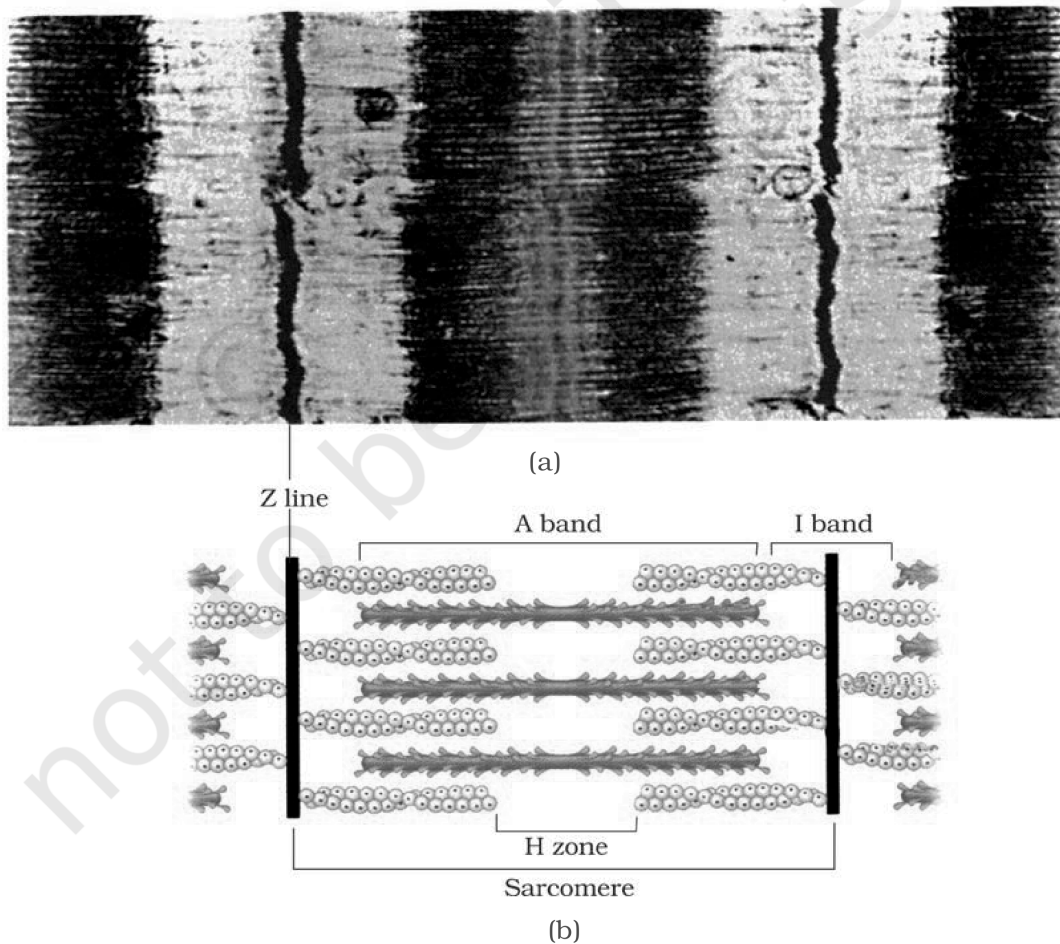


Figure 17.2 Diagrammatic representation of (a) anatomy of a muscle fibre showing a sarcomere (b) a sarcomere

① **[NOTE]** Figure 17.2 labels: Z line; A band; I band; H zone; Sarcomere; panels (a) and (b).

17.2.1 Structure of Contractile Proteins

Actin (thin) filament. Each actin filament is made of two '**F**' (**filamentous**) actins helically wound to each other. Each 'F' actin is a polymer of monomeric '**G**' (**Globular**) actins.

- Two filaments of another protein, **tropomyosin**, also run close to the 'F' actins throughout their length.
- A complex protein **Troponin** is distributed at regular intervals on the tropomyosin. In the **resting state**, a subunit of troponin **masks the active binding sites for myosin** on the actin filaments (Figure 17.3a).

Myosin (thick) filament. Each myosin filament is also a polymerised protein. Many monomeric proteins called **Meromyosins** (Figure 17.3b) constitute one thick filament.

Part of a meromyosin	Name	Feature
Globular head + short arm	Heavy meromyosin (HMM)	Projects outwards at regular distance and angle from the surface of the filament, forming the cross arm ; the head is an active ATPase enzyme with binding sites for ATP and active sites for actin .
Tail	Light meromyosin (LMM)	The rod-like shaft of the monomer.

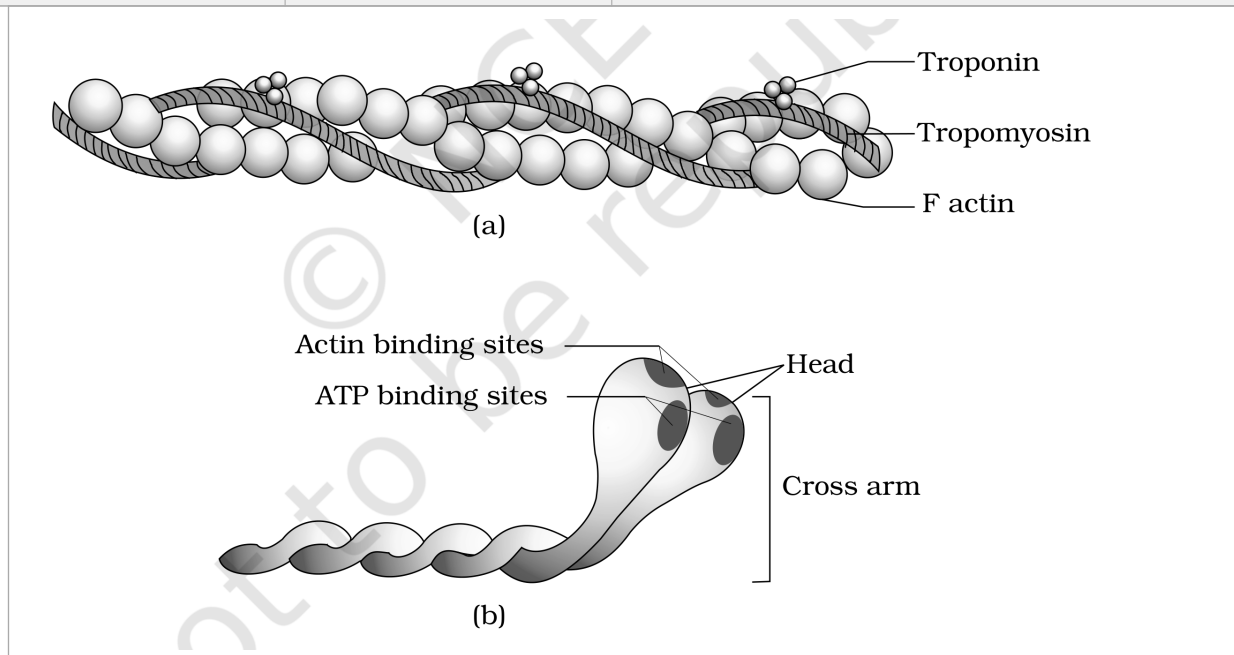


Figure 17.3 (a) An actin (thin) filament (b) Myosin monomer (Meromyosin)

① [NOTE] Figure 17.3 labels: Troponin; Tropomyosin; F actin; Actin binding sites; ATP binding sites; Head; Cross arm; panels (a) and (b).

17.2.2 Mechanism of Muscle Contraction

- **Sliding filament theory** - the mechanism of muscle contraction is best explained by this theory, which states that contraction of a muscle fibre takes place by the **sliding of the thin filaments over the thick filaments**.
- **Motor unit** - a motor neuron along with the muscle fibres connected to it. The junction between a motor neuron and the sarcolemma of the muscle fibre is the **neuromuscular junction** or **motor-end plate**.

The contraction cycle, step by step:

- 1 Contraction is initiated by a signal sent by the **central nervous system (CNS)** via a **motor neuron**.
- 2 The neural signal reaching the neuromuscular junction releases a neurotransmitter (**Acetyl choline**), which generates an **action potential** in the sarcolemma.
- 3 The action potential spreads through the muscle fibre and causes the release of **calcium ions (Ca^{++})** into the sarcoplasm.
- 4 The rise in Ca^{++} level leads to the binding of calcium with a subunit of **troponin** on the actin filaments, which **removes the masking** of the active sites for myosin.
- 5 Using energy from **ATP hydrolysis**, the **myosin head** binds to the exposed active sites on actin to form a **cross bridge** (Figure 17.4).
- 6 The cross bridge **pulls the attached actin filaments towards the centre of the 'A' band**; the 'Z' lines attached to these actins are also pulled inwards, **shortening the sarcomere** - i.e. contraction.
- 7 The myosin head, releasing the **ADP and P_i** , goes back to its relaxed state. A **new ATP** binds and the cross bridge is **broken** (Figure 17.4).

8

The ATP is again hydrolysed by the myosin head and the cycle of cross-bridge formation and breakage repeats, causing further **sliding**.

9

The process continues till the Ca^{++} ions are **pumped back** into the sarcoplasmic cisternae, masking the actin filaments again; the 'Z' lines return to their original position - i.e. **relaxation**.

- During shortening (contraction), the '**I**' bands **get reduced**, whereas the '**A**' bands **retain their length** (Figure 17.5). The figure stages this across three states of the sarcomere: **Relaxed**, **Contracting** and **Maximally Contracted**.
- The **reaction time** of the fibres can vary in different muscles.

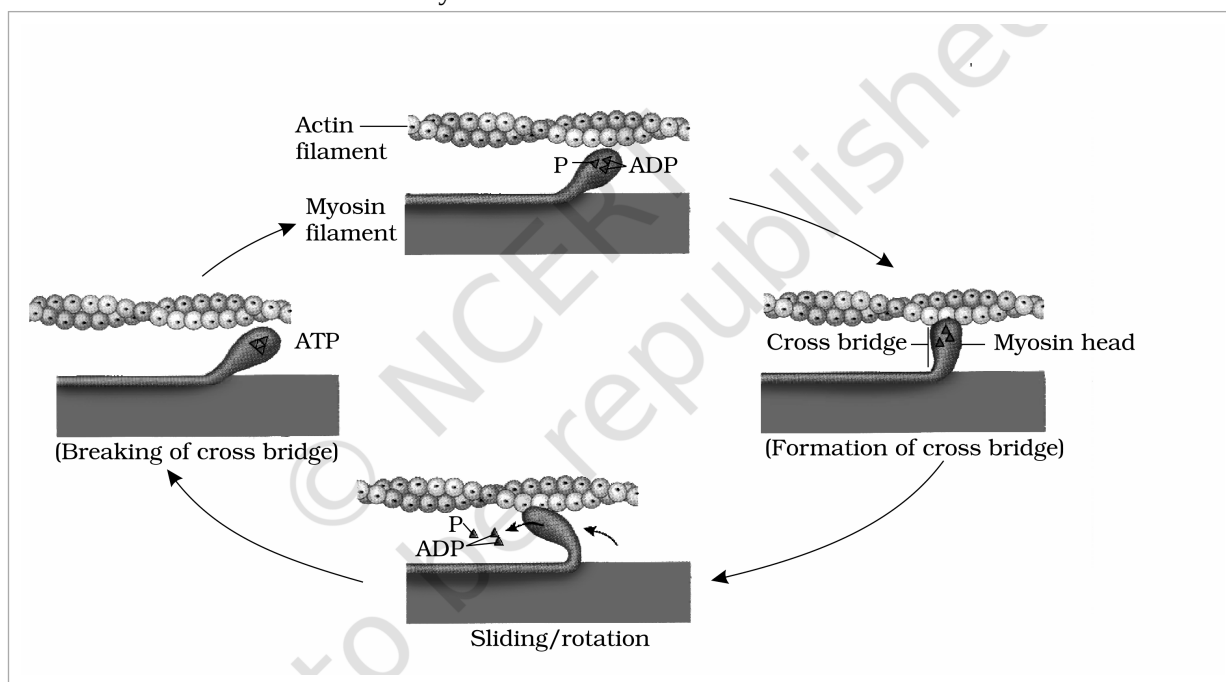


Figure 17.4 Stages in cross bridge formation, rotation of head and breaking of cross bridge

① [NOTE] Figure 17.4 labels: Actin filament; Myosin filament; P; ADP; ATP; Cross bridge; Myosin head; Sliding/rotation of the head; (Formation of cross bridge); (Breaking of cross bridge).

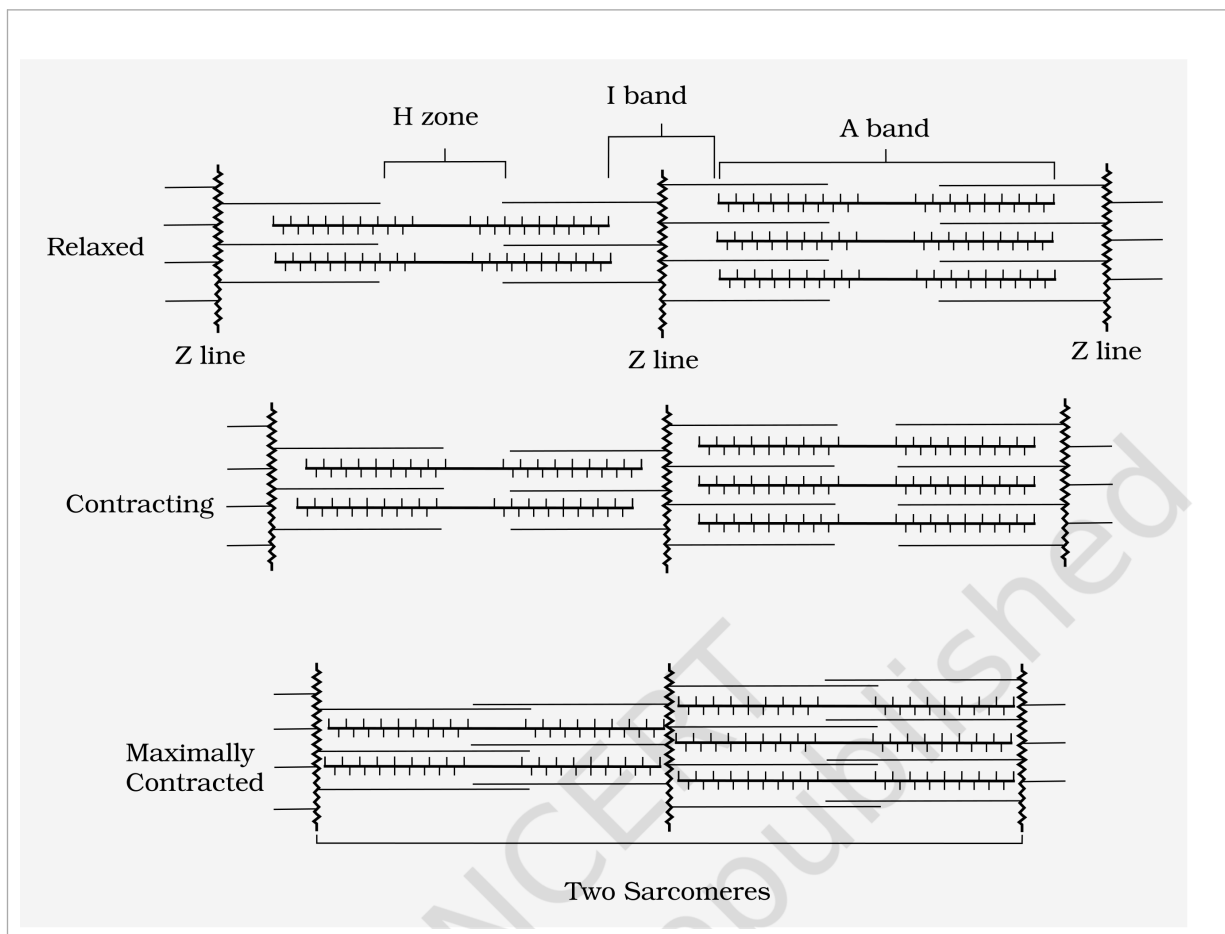


Figure 17.5 Sliding-filament theory of muscle contraction (movement of the thin filaments and the relative size of the I band and H zones)

① **[NOTE]** Figure 17.5 labels: H zone; I band; A band; Z line; Two Sarcomeres; and the three staged states Relaxed, Contracting and Maximally Contracted.

Fatigue. Repeated activation of the muscles can lead to the accumulation of **lactic acid** due to the **anaerobic breakdown of glycogen** in them, causing **fatigue**.

Red vs White fibres. Muscles are classified as Red and White fibres based primarily on the amount of the red-coloured oxygen-storing pigment **myoglobin**.

Feature	Red fibres	White fibres
Myoglobin	High - gives a reddish appearance	Low - appear pale / whitish
Mitochondria	Plenty	Few
Sarcoplasmic reticulum	Lower	High
Energy source	Use stored O_2 for ATP - aerobic muscles	Depend on the anaerobic process for energy

17.3

Skeletal System

The **skeletal system** consists of a framework of **bones and a few cartilages**. This system has a significant role in the movement shown by the body - imagine chewing food without jaw bones and walking around without limb bones.

● **Bone and cartilage** are specialised **connective tissues**. Bone has a very **hard matrix** due to **calcium salts**; cartilage has a slightly **pliable matrix** due to **chondroitin salts**.

- In human beings, the skeletal system is made up of **206 bones** and a few cartilages, grouped into two principal divisions - the **axial** and the **appendicular** skeleton.

The **axial skeleton** comprises **80 bones** distributed along the main axis of the body: the **skull, vertebral column, sternum and ribs**.

Skull (Figure 17.6). The skull is composed of two sets of bones - **cranial and facial** - that total **22 bones**. Cranial bones are **8** in number and form the hard protective outer covering, the **cranium**, for the brain. The facial region is made up of **14** skeletal elements that form the front part of the skull.

- A single **U-shaped bone called hyoid** is present at the base of the buccal cavity.
- Each middle ear contains **three tiny bones - Malleus, Incus and Stapes**, collectively called **Ear Ossicles**.
- The skull articulates with the superior region of the vertebral column with the help of **two occipital condyles (dicondylic skull)**.

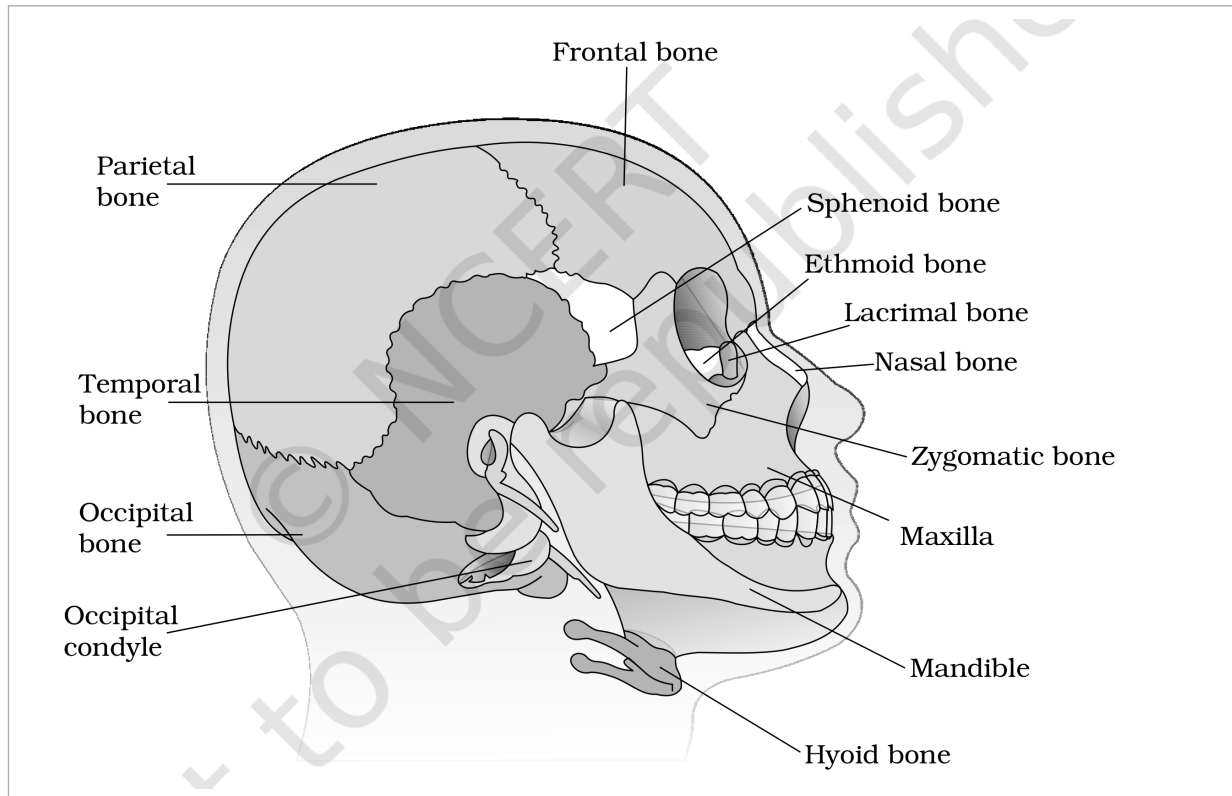


Figure 17.6 Diagrammatic view of human skull

① **[NOTE] Figure 17.6 labels:** Parietal bone; Frontal bone; Temporal bone; Occipital bone; Occipital condyle; Sphenoid bone; Ethmoid bone; Lacrimal bone; Nasal bone; Zygomatic bone; Maxilla; Mandible; Hyoid bone.

Vertebral column (Figure 17.7) is formed by **26 serially arranged units called vertebrae** and is dorsally placed. It extends from the base of the skull and constitutes the main framework of the trunk.

- Each vertebra has a central hollow portion (**neural canal**) through which the **spinal cord** passes. The first vertebra is the **atlas**, and it articulates with the occipital condyles.
- The column is differentiated into **cervical (7), thoracic (12), lumbar (5), sacral (1-fused) and coccygeal (1-fused)** regions, starting from the skull. The number of **cervical vertebrae is seven in almost all mammals**, including human beings. (Adjacent vertebrae are separated by an **intervertebral disc** of cartilage, which forms the cartilaginous joint discussed in 17.4.)
- The vertebral column **protects the spinal cord, supports the head**, and serves as the point of attachment for the ribs and the musculature of the back.

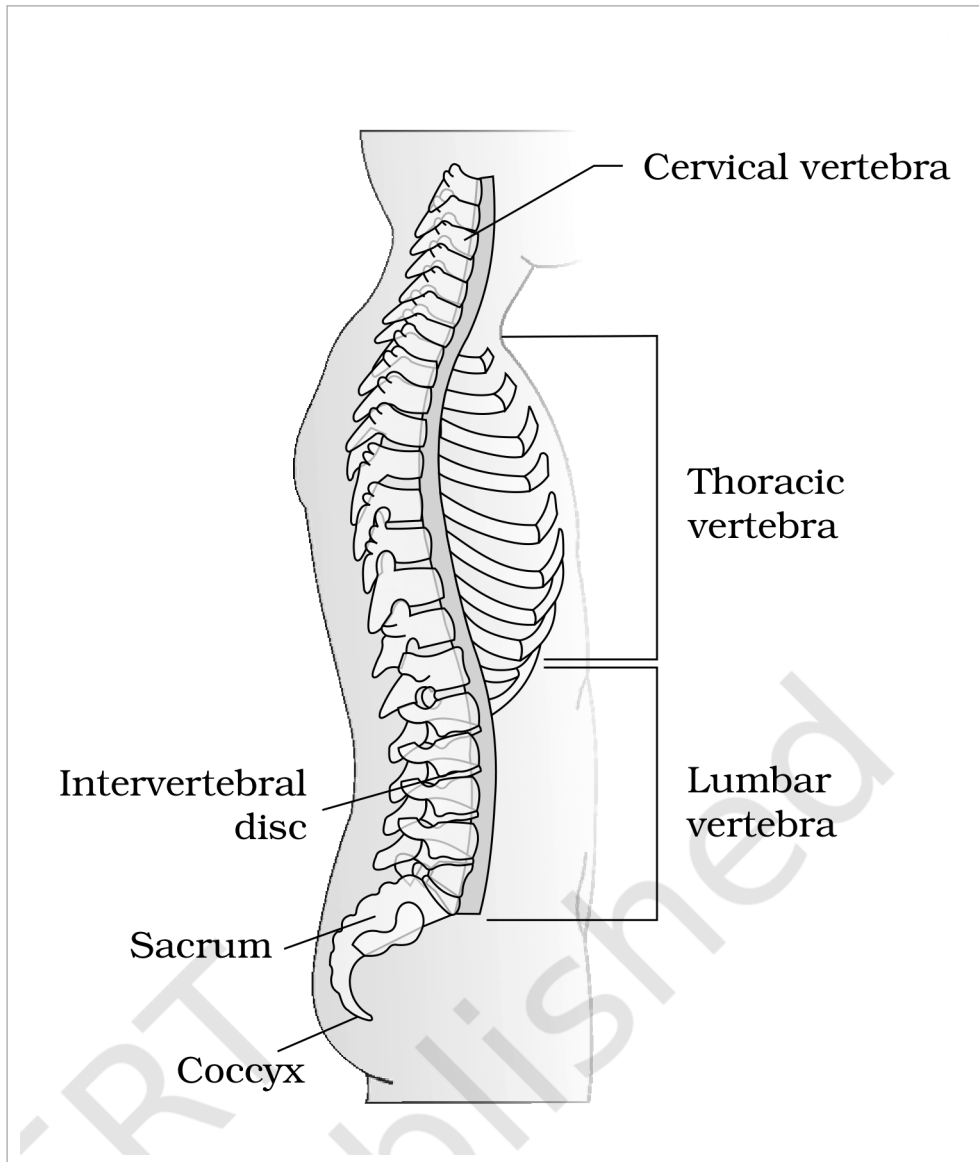


Figure 17.7 Vertebral column (right lateral view)

① [NOTE] Figure 17.7 labels: Cervical vertebra; Thoracic vertebra; Lumbar vertebra; Intervertebral disc; Sacrum; Coccyx.

Sternum and ribs. The **sternum** is a flat bone on the ventral midline of the thorax. There are **12 pairs of ribs**. Each rib is a thin flat bone connected dorsally to the vertebral column and ventrally to the sternum; it has **two articulation surfaces** on its dorsal end and is hence called **bicephalic**.

Rib set	Pairs	Attachment
True ribs	First 7 pairs	Dorsally attached to the thoracic vertebrae; ventrally connected to the sternum with hyaline cartilage
Vertebrochondral (false) ribs	8th, 9th, 10th pairs	Do not articulate directly with the sternum; join the seventh rib with hyaline cartilage
Floating ribs	Last 2 pairs (11th, 12th)	Not connected ventrally

- Thoracic vertebrae, ribs and sternum together form the **rib cage** (Figure 17.8).

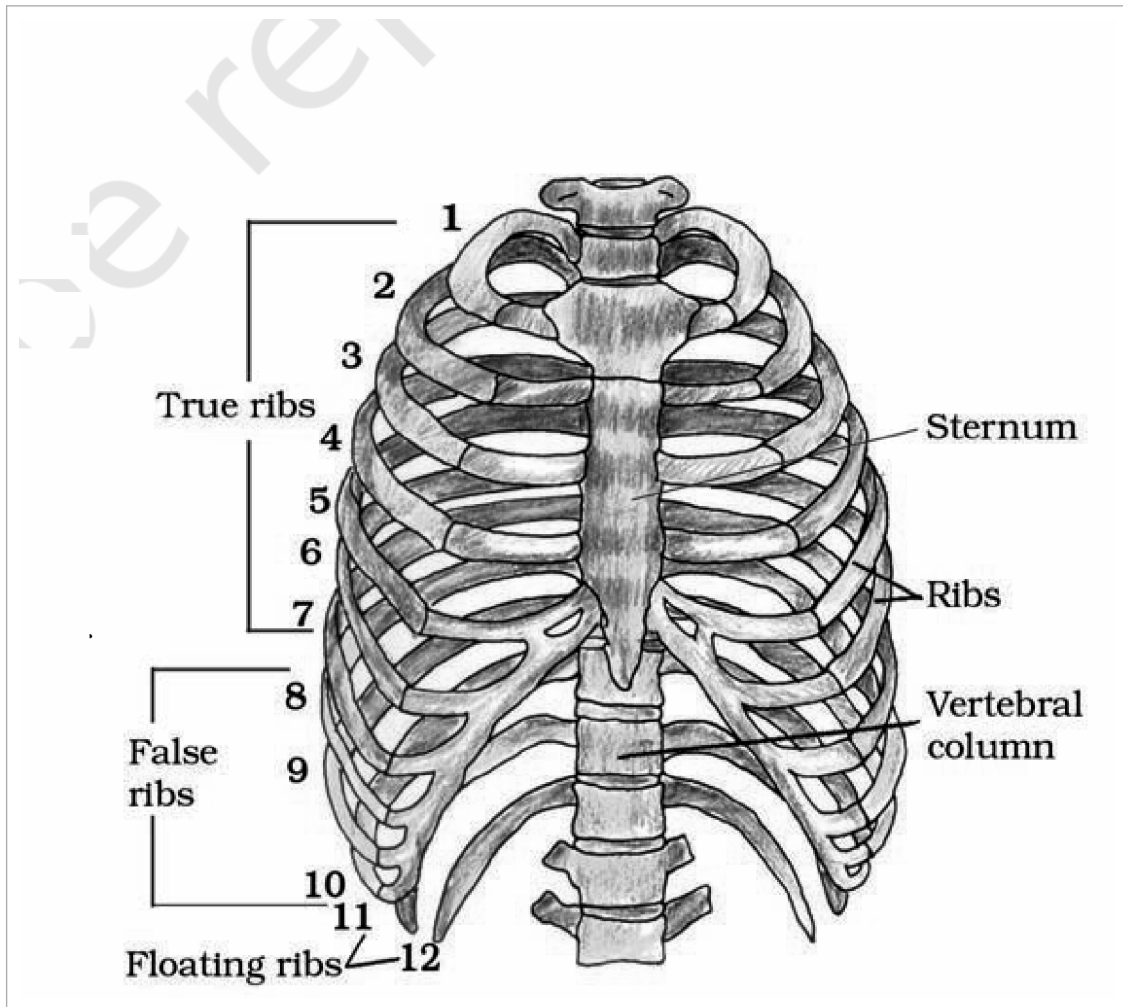


Figure 17.8 Ribs and rib cage

① [NOTE] Figure 17.8 labels: rib pairs numbered 1 to 12; True ribs; False ribs; Floating ribs; Sternum; Ribs; Vertebral column.

17.3.append Appendicular Skeleton

● **Appendicular skeleton** - the bones of the limbs along with their girdles. Each limb is made of **30 bones**.

Limb	Bones (with counts)
Hand / fore limb (Figure 17.9)	Humerus, radius and ulna, carpals (wrist bones - 8), metacarpals (palm bones - 5) and phalanges (digits - 14).
Leg / hind limb (Figure 17.10)	Femur (thigh bone - the longest bone), tibia and fibula, tarsals (ankle bones - 7), metatarsals (5) and phalanges (digits - 14).

- A cup-shaped bone called **patella** covers the knee ventrally (the **knee cap**).
- **Pectoral** and **Pelvic** girdle bones help in the articulation of the upper and lower limbs respectively with the axial skeleton. Each girdle is formed of two halves.

Pectoral girdle (Figure 17.9):

- Each half consists of a **clavicle** and a **scapula**.
- **Scapula** is a large **triangular flat bone** situated in the dorsal part of the thorax between the **second and the seventh ribs**. Its dorsal, flat, triangular body has a slightly elevated ridge called the **spine**, which projects as a flat, expanded process called the **acromion**; the clavicle articulates with this.
- Below the acromion is a depression called the **glenoid cavity**, which articulates with the **head of the humerus** to form the **shoulder joint**.
- Each **clavicle** is a long slender bone with two curvatures, commonly called the **collar bone**.

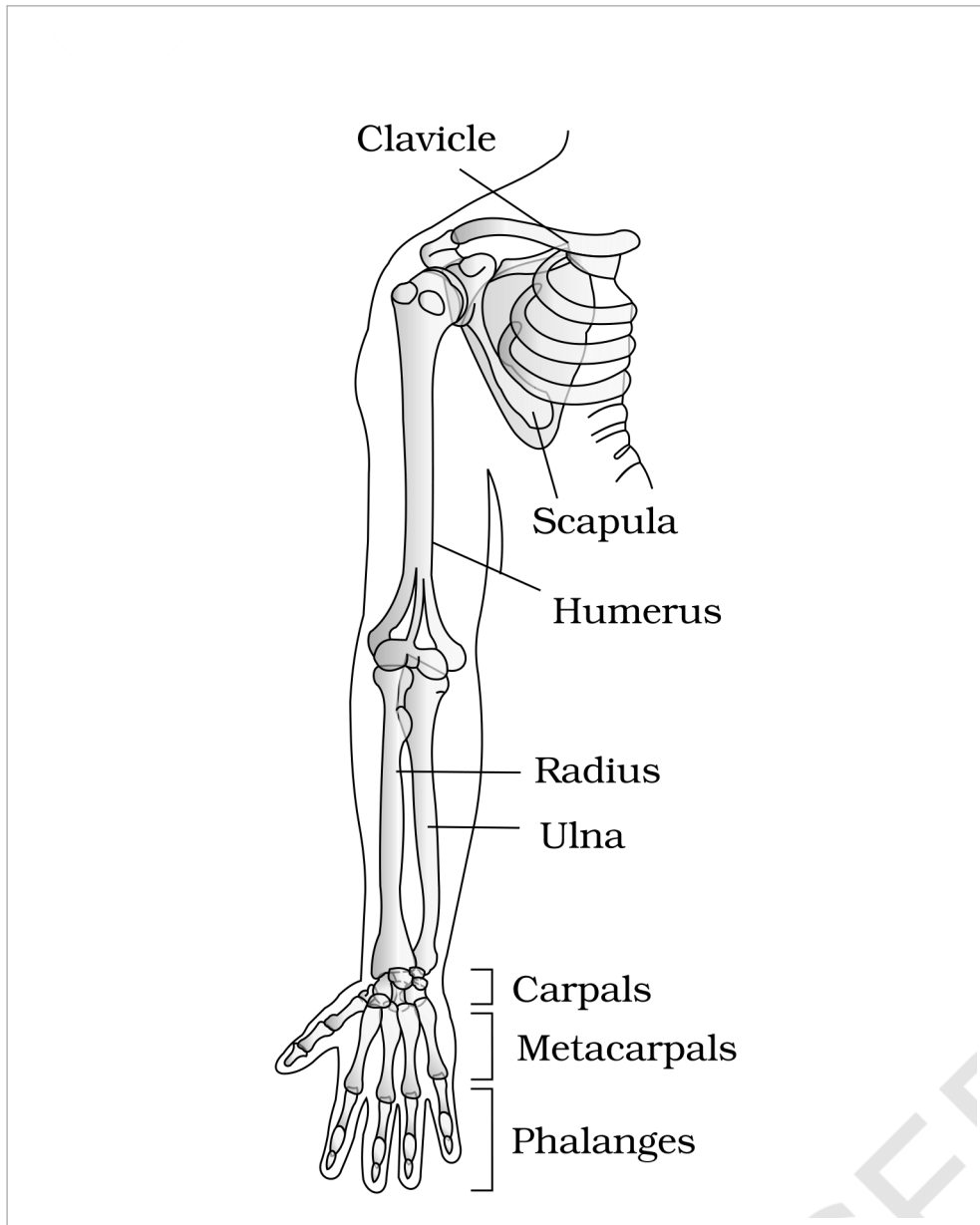


Figure 17.9 Right pectoral girdle and upper arm. (frontal view)

① [NOTE] Figure 17.9 labels: Clavicle; Scapula; Humerus; Radius; Ulna; Carpals; Metacarpals; Phalanges.

Pelvic girdle (Figure 17.10):

- Consists of **two coxal bones**. Each coxal bone is formed by the fusion of **three bones - ilium, ischium and pubis**.
- At the point of fusion is a cavity called the **acetabulum**, to which the thigh bone (femur) articulates.
- The two halves of the pelvic girdle meet ventrally to form the **pubic symphysis**, containing fibrous cartilage.

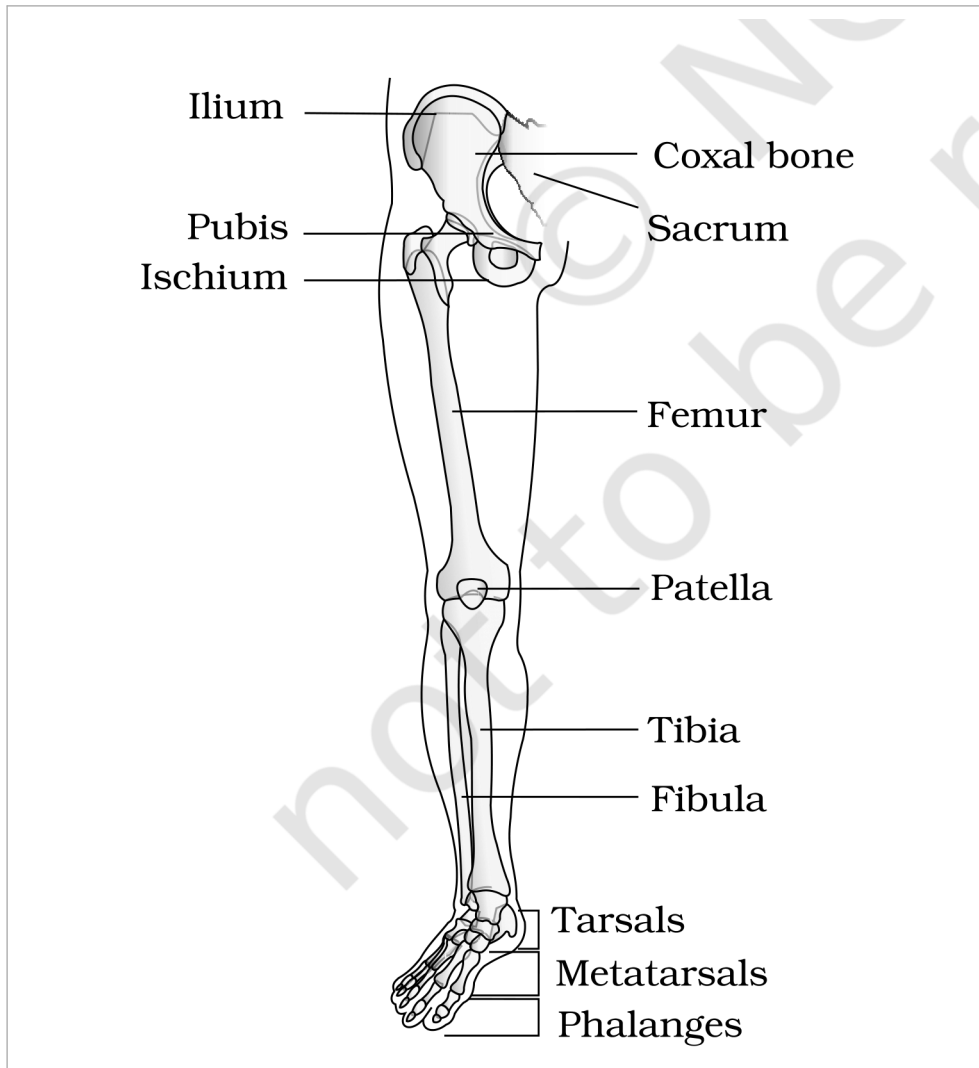


Figure 17.10 Right pelvic girdle and lower limb bones (frontal view)

① [NOTE] **Figure 17.10 labels:** Ilium; Pubis; Ischium; Coxal bone; Sacrum; Femur; Patella; Tibia; Fibula; Tarsals; Metatarsals; Phalanges.

☆ [MEMORY AID - not in NCERT] **Pectoral vs Pelvic girdle** at a glance: pectoral = **clavicle** + **scapula** (articulates the arm at the glenoid cavity/shoulder); pelvic = one **coxal bone** per half = **ilium** + **ischium** + **pubis** fused (articulates the thigh at the acetabulum).

17.4

Joints

Joints are essential for all types of movements involving the bony parts of the body; locomotory movements are no exception.

- **Joints** - points of contact between bones, or between bones and cartilages. The force generated by the muscles is used to carry out movement through joints, where the joint acts as a **fulcrum**. The movability at these joints varies depending on different factors.

Joints have been classified into **three major structural forms**, namely, **fibrous, cartilaginous and synovial**.

Joint type	Movement	Example
Fibrous	Do not allow any movement	Flat skull bones fusing end-to-end via dense fibrous connective tissue as sutures to form the cranium
Cartilaginous	Permit limited movements	Joint between adjacent vertebrae in the vertebral column
Synovial	Allow considerable movement; help in locomotion and many other movements	Characterised by a fluid-filled synovial cavity between the articulating surfaces of the two bones

Synovial joints are characterised by the presence of a **fluid-filled synovial cavity** between the articulating surfaces of the two bones. Such an **arrangement** allows considerable movement. Sub-types with examples:

- **Ball and socket joint** - between humerus and pectoral girdle.
- **Hinge joint** - knee joint.
- **Pivot joint** - between atlas and axis.
- **Gliding joint** - between the carpals.
- **Saddle joint** - between carpal and metacarpal of the thumb.

17.5

Disorders of Muscular and Skeletal System

Disorder	Description
Myasthenia gravis	Auto immune disorder affecting the neuromuscular junction , leading to fatigue, weakening and paralysis of skeletal muscle.
Muscular dystrophy	Progressive degeneration of skeletal muscle, mostly due to a genetic disorder .
Tetany	Rapid spasms (wild contractions) in muscle due to low Ca^{++} in body fluid.
Arthritis	Inflammation of joints.
Osteoporosis	Age-related disorder characterised by decreased bone mass and increased chances of fractures. Decreased levels of estrogen is a common cause.
Gout	Inflammation of joints due to the accumulation of uric acid crystals .

① **[NOTE]** Chapter map recap: **movement** (amoeboid / ciliary / muscular) leads to **muscle** (skeletal / visceral / cardiac), whose contractile unit is the **sarcomere**; contraction runs by the **sliding filament theory**; muscles act on the **skeletal system** (206 bones: axial + appendicular) through **joints**. Disorders can strike either the muscular side (myasthenia gravis, muscular dystrophy, tetany) or the skeletal/joint side (arthritis, osteoporosis, gout).